

| >> Rational Functions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Unit | CHECKPOINT |   |   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------------|---|---|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |      | 1          | 2 | 3 |
| <p><b>2A.6 Cubic, cube root, absolute value and rational functions, equations, and inequalities.</b> The student applies mathematical processes to understand that cubic, cube root, absolute value and rational functions, equations, and inequalities can be used to model situations, solve problems, and make predictions.</p> <p><b>2A.7 Number and algebraic method.</b> The student applies mathematical processes to understand that exponential and logarithmic functions can be used to model situations and solve problems.</p> <p style="text-align: right;">Connected Knowledge and Skills 2A.2</p> |      |            |   |   |

| Process (Tools to Know)                                                                                                                            | Unit | CHECKPOINT |   |   |
|----------------------------------------------------------------------------------------------------------------------------------------------------|------|------------|---|---|
|                                                                                                                                                    |      | 1          | 2 | 3 |
| <p>2A.1(A) apply math in everyday situations ③</p> <p>2A.1(B) use problem-solving models ③</p> <p style="text-align: right;">connected 2A.1(C)</p> |      |            |   |   |

| Content                                                                                                                                                                                                                                                                                                                                  | Unit | CHECKPOINT                                  |   |   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---------------------------------------------|---|---|
|                                                                                                                                                                                                                                                                                                                                          |      | 1                                           | 2 | 3 |
| <b>Describing Rational Functions</b>                                                                                                                                                                                                                                                                                                     |      |                                             |   |   |
| 2A.2(A) graph the functions $f(x)=\sqrt{x}$ , $f(x)=1/x$ , $f(x)=x^3$ , $f(x)=\sqrt[3]{x}$ , $f(x)=b^x$ , $f(x)= x $ , and $f(x)=\log_b(x)$ where $b$ is 2, 10, and $e$ , and, when applicable, analyze the key attributes such as domain, range, intercepts, symmetries, asymptotic behavior, and maximum and minimum given an interval |      | Data included in "Absolute Value Functions" |   |   |
| 2A.6(G) analyze the effect on the graphs of $f(x) = 1/x$ when $f(x)$ is replaced by $af(x)$ , $f(bx)$ , $f(x-c)$ , and $f(x) + d$ for specific positive and negative real values of $a$ , $b$ , $c$ , and $d$                                                                                                                            |      |                                             |   |   |
| 2A.6(K) determine the asymptotic restrictions on the domain of a rational function and represent domain and range using interval notation, inequalities, and set notation                                                                                                                                                                |      |                                             |   |   |

|                                                                                                                                                |  |                                                                     |  |  |
|------------------------------------------------------------------------------------------------------------------------------------------------|--|---------------------------------------------------------------------|--|--|
| <b>Simplifying and Solving Rational Equations</b>                                                                                              |  |                                                                     |  |  |
| 2A.6(I) solve rational equations that have real solutions ③                                                                                    |  |                                                                     |  |  |
| 2A.6(L) formulate and solve equations involving inverse variation                                                                              |  |                                                                     |  |  |
| 2A.7(F) determine the sum, difference, product, and quotient of rational expressions with integral exponents of degree one and of degree two   |  |                                                                     |  |  |
| 2A.6(H) formulate rational equations that model real-world situations ③                                                                        |  |                                                                     |  |  |
| 2A.6(J) determine the reasonableness of a solution to a rational equation                                                                      |  |                                                                     |  |  |
| 2A.7(C) determine the quotient of a polynomial of degree three and of degree four when divided by a polynomial of degree one and of degree two |  | Data included in "Cube Root, Cubic, and Other Polynomial Functions" |  |  |

| Process (Ways to Show)                                                                                                                  | Unit | CHECKPOINT |   |   |
|-----------------------------------------------------------------------------------------------------------------------------------------|------|------------|---|---|
|                                                                                                                                         |      | 1          | 2 | 3 |
| <p>2A.1(E) create representations</p> <p>2A.1(F) analyze information ③</p> <p style="text-align: right;">connected 2A.1(D), 2A.1(G)</p> |      |            |   |   |

>> TEKS clusters typically requiring additional time and focus in the curriculum

| PROCESS STANDARDS: MATHEMATICAL PROCESS STANDARDS |                                                                                                | Unit          | CHECKPOINT |   |   |
|---------------------------------------------------|------------------------------------------------------------------------------------------------|---------------|------------|---|---|
|                                                   |                                                                                                |               | 1          | 2 | 3 |
| 2A.1                                              | The student uses mathematical processes to acquire and demonstrate mathematical understanding. | Tools to Know |            |   |   |
|                                                   |                                                                                                | Ways to Show  |            |   |   |

| TOOLS TO KNOW |                                                                                                                                                                                                                                                 | Unit | CHECKPOINT |   |   |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------------|---|---|
|               |                                                                                                                                                                                                                                                 |      | 1          | 2 | 3 |
| 2A.1(A)       | apply mathematics to problems arising in everyday life, society, and the workplace ⑧                                                                                                                                                            |      |            |   |   |
| 2A.1(B)       | use a problem-solving model that incorporates analyzing given information, formulating a plan or strategy, determining a solution, justifying the solution, and evaluating the problem-solving process and the reasonableness of the solution ⑧ |      |            |   |   |
| 2A.1(C)       | select tools, including real objects, manipulatives, paper and pencil, and technology as appropriate, and techniques, including mental math, estimation, and number sense as appropriate, to solve problems                                     |      |            |   |   |

| WAYS TO SHOW |                                                                                                                                                                    | Unit | CHECKPOINT |  |  |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------------|--|--|
|              |                                                                                                                                                                    |      |            |  |  |
| 2A.1(D)      | communicate mathematical ideas, reasoning, and their implications using multiple representations, including symbols, diagrams, graphs, and language as appropriate |      |            |  |  |
| 2A.1(E)      | create and use representations to organize, record, and communicate mathematical ideas                                                                             |      |            |  |  |
| 2A.1(F)      | analyze mathematical relationships to connect and communicate mathematical ideas ⑧                                                                                 |      |            |  |  |
| 2A.1(G)      | display, explain, and justify mathematical ideas and arguments using precise mathematical language in written or oral communication                                |      |            |  |  |